

Study Questions: Action — Modifying the Pattern

Mid-Row

1. Explain why every stitch is an "action" in the Active Inference sense. What is each stitch doing to the environment?
2. When you substitute half double crochet for double crochet, what part of the generative model are you modifying? What downstream effects would you predict?
3. Describe the two ways Active Inference agents can reduce prediction error. Give a crochet example of each.
4. Why does adjusting a pattern for a different size require understanding the "deep structure" of the pattern rather than just changing numbers?
5. A crocheter discovers a missed stitch 3 rows back. They must decide: frog, tink, or leave it. What factors should inform this decision in terms of precision weighting?
6. How does improvisation (freeform crochet without a pattern) differ from pattern-following in terms of the perception-action loop?
7. A pattern calls for a specific yarn that has been discontinued. How is choosing a substitute yarn an act of active inference?
8. Explain why the decision to "leave an error" is a legitimate active inference strategy, not just laziness. Under what conditions is it optimal?
9. How does modifying a pattern for a different yarn weight (e.g., converting from worsted to bulky) require re-parameterizing the generative model?
10. When you add an unplanned border to a finished piece, you are extending the generative model beyond its original scope. What predictions must your model generate for this new addition?

11. Describe a situation where perceptual inference (updating your expectations) is more appropriate than active inference (changing the fabric). When should you adjust your model rather than your work?
12. How does the cost of frogging change with the size of the project? Why does this matter for the precision-weighted error correction decision?
13. A crocheter looks at their gauge swatch and decides to go up a hook size. Is this active inference or model updating? Explain.
14. How does color substitution in a colorwork pattern differ from stitch substitution in terms of the generative model changes involved?
15. When two crocheters make the same pattern with different modifications, they are exploring different regions of the model's parameter space. What does this mean?
16. Explain why experienced crocheters are better at improvisation. What properties of their generative model enable it?
17. A crocheter is making a garment that does not fit as expected. They can modify the remaining construction or accept the current trajectory. Frame this as a planning decision under active inference.
18. How does the concept of "design ease" (extra room built into garment patterns) relate to the pattern designer's model of acceptable prediction error?
19. When you "wing it" on a finishing technique (picking a border style on the fly), what is your generative model doing?
20. Describe the difference between a planned modification (decided before starting) and a reactive modification (decided after discovering a problem). How does each relate to active inference?